ML Model Deployment Strategies and Model Serving using FastAPI

Dataset: Titanic – Machine Learning from Disaster (Kaggle train.csv)

Objective: Train and evaluate a classification pipeline, then reuse the same saved model for online, batch, streaming, and REST API inference.

Observations

1. Online Inference vs Batch Inference
Online inference predicts one passenger record at a time and is useful when an immediate result is needed. Batch inference predicts a group of records together and is useful for bulk processing.

2. Streaming Inference vs Batch Inference
Streaming inference handles records continuously as they arrive; this assignment simulates it by processing one record every second. Batch inference collects a group of records and predicts them together.

3. Why FastAPI is used for model serving
FastAPI exposes the trained model through REST endpoints. Pydantic validates incoming data, and Swagger UI at /docs supports interactive testing and documentation.

4. Why the same trained model should be reused
Reusing model.pkl keeps preprocessing and classification consistent across all inference strategies. Retraining separately could lead to different predictions and make comparisons unfair.

Accuracy reporting

The measured accuracy is printed by train_model.py and saved in reports/evaluation.txt after train.csv is placed in data/. The accuracy is intentionally not hard-coded in this report because it must match the actual held-out split and execution.

Dataset link: https://www.kaggle.com/competitions/titanic/data